

Phase III Randomized Trial of Cisplatin Plus Placebo Compared With Cisplatin Plus Cetuximab in Metastatic/Recurrent Head and Neck Cancer: An Eastern Cooperative Oncology Group Study

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Submitted April 29, 2005; accepted August 18, 2005.

Supported in part by Public Health Service Grants No. CA23318, CA66636, CA21115, CA07190, and CA16116 from the National Cancer Institute, National Institutes of Health, and the United States Department of Health and Human Services. Its contents are solely the responsibility of the authors and do not necessarily represent the official views of the National Cancer Institute.

Presented in part at the 38th Annual Meeting of the American Society of Clinical Oncology, May 18-21, Orlando, FL, and the American Association for Cancer Research/National Cancer Institute/European Organisation for Research and Treatment of Cancer International Conference on Molecular Targets in Cancer Therapy, November 17-21, 2003, Boston, MA.

Authors' disclosures of potential conflicts of interest are found at the end of this article.

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0732-183X/05/2334-8646/\$20.00

DOI: 10.1200/JCO.2005.02.4646

A B S T R A C T

Purpose

Therapy of recurrent/metastatic squamous cell carcinoma of the head and neck results in median progression-free survival (PFS) of 2 months. These cancers are rich in epidermal growth factor receptor (EGFR). We wished to determine whether the addition of cetuximab, which inhibits activation of EGFR, would improve PFS.

Patients and Methods

Patients with recurrent/metastatic squamous cell carcinoma of the head and neck were randomly assigned to receive cisplatin every 4 weeks, with weekly cetuximab (arm A) or placebo (arm B). Tumor tissue was assayed for EGFR expression by immunohistochemistry. The primary end point was PFS. Secondary end points of interest were response rate, toxicity, overall survival, and correlation of EGFR with clinical end points.

Results

There were 117 analyzable patients enrolled. Median PFS was 2.7 months for arm B and 4.2 months for arm A. The hazard ratio for progression of arm A to arm B was 0.78 (95% CI, 0.54 to 1.12). Median overall survival was 8.0 months for arm B and 9.2 months for arm A ($P = .21$). The hazard ratio for survival by skin toxicity in cetuximab-treated patients was 0.42 (95% CI, 0.21 to 0.86). Objective response rate was 26% for arm A and 10% for arm B ($P = .03$). Enhancement of response was greater for patients with EGFR staining present in less than 80% of cells.

Conclusion

Addition of cetuximab to cisplatin significantly improves response rate. There was a survival advantage for the development of rash. Progression-free and overall survival were not significantly improved by the addition of cetuximab in this study.

J Clin Oncol 23:8646-8654. © 2005 by American Society of Clinical Oncology

INTRODUCTION

Squamous cell carcinoma of the head and neck remains a challenging clinical problem, with 29,370 new cases per year in the United States and half a million new cases annually worldwide.¹ Successful management often leaves patients with significant debility, and patients with advanced disease are unlikely to be cured by current therapies. Management after recurrence has relied on the use of systemic therapy.

Platinum-based chemotherapy has been the standard, and although higher response rates are observed with combinations, no survival advantage has been demonstrated for any regimen over cisplatin monotherapy.²⁻⁴ The median progression-free survival for patients with recurrent or metastatic disease and Eastern Cooperative Oncology Group (ECOG) performance status of 0 to 2 is reported to be 2 months.² Novel systemic treatments are urgently needed for these patients.

The epidermal growth factor receptor (EGFR) is commonly expressed in squamous cell carcinoma of the head and neck. The level of expression may be higher in the tumor than in the adjacent normal tissue, and this relative overexpression is associated with poor prognosis.^{5,6} EGFR expression has been measured by a variety of techniques, of which the most widely applied is immunohistochemistry. Cetuximab (C225, Erbitux; Imclone Systems Inc, Branchburg, NJ) is a monoclonal antibody that binds the extracellular portion of the EGFR and interferes with binding and receptor activation by the natural ligands of EGFR. This binding is of high affinity.⁷ Phase I studies demonstrated the safety of this agent given alone or in combination with chemotherapy.⁸ The recommended phase II dose was that dose at which clearance was saturated. This dose achieves partial inhibition of EGFR phosphorylation in a proportion of patients.⁹ The principal toxicities of cetuximab at the recommended dose and schedule are hypersensitivity reactions and acneiform rash. Biopsy of the rash demonstrates neutrophilic folliculitis.¹⁰

Activity in head and neck cancer was demonstrated when cetuximab was given in combination with cisplatin chemotherapy, including in patients with documented disease progression during cisplatin therapy.⁹ We wished to determine whether the addition of cetuximab to standard cisplatin monotherapy would improve progression-free survival. Secondary end points of interest were response rate, toxicity, overall survival (OS), and correlation of EGFR with clinical end points.

PATIENTS AND METHODS

Eligibility

Patients were eligible if they had measurable or nonmeasurable but assessable squamous cell carcinoma of the head and neck region, which was recurrent after locoregional therapy or metastatic. If the only site of measurable disease was a previously irradiated area, documented progression of disease and a 4-week period since the conclusion of radiotherapy or biopsy-proven residual disease at least 8 weeks from the completion of radiation therapy were required. No induction or adjuvant chemotherapy within 3 months of study entry, or prior chemotherapy for recurrent or metastatic disease was permitted, and patients must have recovered from the effects of any major surgery. Patients were required to be more than 2 years disease free from any prior malignancy, except curatively treated basal or squamous cell carcinoma of the skin or carcinoma-in-situ of the cervix. They might not have had evidence of active infection or been receiving treatment for infection at the time of enrollment.

Patients were required to have ECOG performance status of 0 or 1, to be ≥ 18 years of age, and to be without history of brain metastases or evidence of hypersensitivity to murine proteins. Adequate organ function was required, manifest by absolute neutrophil count $\geq 1,500/\text{mL}$, platelet count $\geq 100,000/\text{mL}$, hemoglobin $\geq 10 \text{ gm/dL}$, serum creatinine $\leq 1.2 \text{ mg\%}$ or creatinine clearance $\geq 50 \text{ mL/min}$, total bilirubin $\leq 1.5 \text{ mg/100 mL}$, and

AST, ALT, and alkaline phosphatase ≤ 2 times the institution's upper limit of normal. Pregnant and lactating women were excluded, and women of childbearing potential must have had a negative serum pregnancy test. All patients were required to provide written informed consent. This consent form was approved by local investigational review boards.

Treatment Assignment

Patients were equally randomly assigned to cisplatin with placebo and cisplatin with cetuximab regimens. The treatments were assigned using permuted blocks within strata with dynamic balancing within main institutions and their affiliate networks. The stratification factors used were disease status (two levels, previously untreated/newly diagnosed and recurrent) and ECOG performance status (two levels, 0 and 1).

Measurement of EGFR Expression

EGFR expression was determined by immunohistochemistry performed at Impath Laboratories (now Genzyme, Westborough, MA), using the DAKO EGFR kit (DAKO Cytomation; Glostrup, Denmark). Unstained slides were prepared from formalin-fixed, paraffin-embedded tissue, either at the treating sites or the ECOG Pathology Coordinating Office, Northwestern University. Cell line control slides containing positive and negative cells were used as controls. If no cell membranes were visualized in the clinical specimen, the assay was repeated with a shorter period of exposure to protease during antigen retrieval, or the assay was repeated on another slide.

EGFR expression was reported by the maximal intensity of the immunohistochemical stain in the cytoplasm on an ordinal scale of 0 to 3. The density of staining was also reported as the proportion of cells staining at that intensity, in increments of 10%. High EGFR expression was defined as 3+ staining on at least 80% of cells. All other degrees of staining are combined, and this group is referred to as low-to-moderate staining, although it should be noted that the majority of these cases displayed high-intensity staining but on a smaller proportion of cells.

Chemotherapy

The treatments to be compared were administered as follows:

Arm A: cetuximab plus cisplatin. Cetuximab was supplied as a 2-mg/mL solution in blinded vials. A dose of 200 mL/m² was given intravenously (IV) on day 1 over 120 minutes for cycle 1 only. Subsequent cycles were administered 125 mL/m²/wk IV over 60 minutes. Cisplatin 100 mg/m² IV was given on day 1 every 4 weeks.

Arm B: placebo plus cisplatin. Placebo was supplied in blinded vials. A dose of 200 mL/m² was given IV on day 1 over 120 minutes for cycle 1 only. Subsequent cycles administered were 125 mL/m²/wk IV over 60 minutes. Cisplatin 100 mg/m² IV was given on day 1 every 4 weeks.

Dose Modifications

Dose modifications for both arms were specified in the protocol. These included substitution of carboplatin for cisplatin for patients who developed grade 2 neurotoxicity or worsening of the creatinine clearance to less than 50 mL/min. Study agent was held for grade 3 skin toxicity until all skin toxicity resolved to grade ≤ 2 , for up to 2 consecutive weeks. If grade 3 skin toxicity recurred, the study agent dose was held and then resumed once skin toxicity was $\leq$ grade 2, with a 25 mL/m² dose reduction. If more than two dose reductions were required, the cetuximab/placebo study agent was discontinued.

Discontinuation of Therapy

Treatment could be discontinued two cycles after complete response (CR) was achieved, at the discretion of the treating physician. Patients achieving partial response (PR) stayed on treatment until a CR was achieved, or there was disease progression. Those with stable disease could discontinue treatment after six cycles. Patients with progression of disease discontinued therapy; however, after October 2000, when amendment 2 was activated, treatment assignments could be unblinded on progression of disease, and patients assigned to arm B were permitted to cross-over to cetuximab therapy if they continued to meet eligibility criteria. Any patient or physician might discontinue therapy if it was no longer deemed to be in the best interest of the patient.

Follow-Up

Toxicity was evaluated using the National Cancer Institute Common Toxicity Criteria version 2.0. ECOG Solid Tumor Response Criteria were used to evaluate response.

Statistical Analysis

The primary end point of the study was progression-free survival (PFS) on step 1. PFS is defined as the time from randomization to the first of progressive disease or death from any cause, censored at last contact. The study was designed to detect a difference in median PFS from 2 months with cisplatin plus placebo to 4 months with the experimental arm, with 90% power using a one-sided log-rank test with an overall type I error rate of 0.025. Assuming constant failure rates over time, the study required 98 eligible patients; the target accrual was 114 patients to allow for ineligible patients and to adjust for those who did not receive the intended therapy. The primary analysis excluded ineligible patients.

χ^2 tests or, where cell frequencies were small, Fisher's exact tests were used to compare categorical variables between groups (eg, toxicities). The Wilcoxon rank sum test was used to compare continuous outcomes between the two arms (eg, the number of treatment cycles received). Multivariable logistic regression was used to examine associations between a dichotomous end point (eg, response) and covariates of interest. Survival distributions were estimated using the Kaplan-Meier method,¹¹ and univariate associations were tested using log-rank tests. Cox proportional hazards regression was used to model the survival data on covariates of interest while adjusting for other covariates and testing for interactions among them. Associations between development of skin toxicity and survival end points were evaluated using Cox regression modeling with a time-varying covariate.¹²

For the primary comparison of PFS, the nominal one-sided log-rank *P* value was compared with a .024 significance level to adjust for one interim analysis, as specified by the study design. OS, defined as time from randomization to death or censored at last contact, is also reported as a one-sided *P* value, but compared with a .025 significance level. All other reported *P* values are two-sided and were compared with a .05 significance level.

RESULTS

A total of 123 patients were entered onto the study between June 30, 1999, and June 13, 2001, and 118 were eligible; however, one eligible patient died before randomization. Of the 117 eligible and analyzable cases, 57 patients were on

arm A and 60 patients were on arm B. At the time of final analysis, the median follow-up time among the seven patients censored for OS was 31 months, with a minimum of 16 and a maximum follow-up time for survivors of 47 months. Thirteen eligible placebo patients who experienced disease progression after activation of amendment 2 crossed over to receive cetuximab therapy at step 2.

Patient Characteristics

Patient characteristics are listed in Table 1. The median age of the patients was 60.6 years on the experimental arm and 58.3 years on the control arm. Seventy-two percent of patients on the cetuximab-containing arm and 83% on the placebo-containing arm were male. There were 35 patients (61%) on the cetuximab-containing arm and 41 patients (68%) on the placebo-containing arm who had known distant metastases. More than 90% of patients on each arm had either prior surgery or radiotherapy or both. Thirty percent of patients in the cisplatin and cetuximab arm and 15% of those in the cisplatin and placebo arm had received prior systemic therapy. The groups were well balanced with respect to performance status, weight loss, smoking status, alcohol consumption, tumor differentiation, and primary site.

Treatment Administration

The median number of cycles of therapy was 4.5 for the cetuximab arm and three for the placebo arm (*P* = .02). A median of two cycles of therapy on step 2 was delivered to the 13 eligible patients who received cetuximab after cross-over to step 2. The most common reason for treatment discontinuation was progressive disease (arm A, 70%; arm B, 74%; step 2, 54%). Other reasons for treatment discontinuation were completion of treatment, toxicity, patient preference, and unknown. The mean cumulative cisplatin dose was 399.5 and 307.0 mg/m² in the cetuximab and placebo arms, respectively (*P* = .01). Eight patients in arm A and 12 patients in arm B received carboplatin for at least one cycle. Four of 13 eligible patients on step 2 were receiving carboplatin at the time of cross-over, and an additional patient had carboplatin substituted for cisplatin while on step 2.

Toxicity

The seven patients who never started therapy are excluded from this analysis. The data reported here (Table 2) concern the toxicities reported for the 58 patients per arm who received therapy. Grade 3 or 4 toxicity was observed in 90% of patients on arm A and 73% of patients on arm B (*P* = .02). The principal toxicities observed were those associated with use of cisplatin at 100 mg/m², including fatigue (17% in arm A v 14% in arm B; *P* = .61), nausea (24% v 19%; *P* = .50), vomiting (17% in each arm; *P* = .81), hyponatremia (26% v 28%; *P* = .83), neutropenia (30% v 14%; *P* = .04), and thrombocytopenia (11% v 4%; *P* = .14).

Table 1. Patient Characteristics

	Cisplatin + Cetuximab		Cisplatin + Placebo	
	No. of Patients	%	No. of Patients	%
Total patients	57	48.7	60	51.3
Age, years				
Median		60.6		58.3
Range		39.8-85.7		32.2-83.5
Sex				
Male	41	71.9	50	83.3
Female	16	28.1	10	16.7
Entry performance status				
0	24	42.1	24	40
1	33	57.9	36	60
Weight loss in previous 6 months				
< 5%	32	56.1	31	51.7
≥ 5%	20	35.1	25	41.7
Unknown	5	8.8	4	6.6
Cell differentiability				
Well differentiated	4	7.0	12	20.0
Moderately differentiated	28	49.1	23	38.3
Poorly differentiated	16	28.1	20	33.3
Unknown	9	15.8	5	8.3
Primary site				
Oral cavity	11	19.3	13	21.7
Oropharynx	18	31.6	19	31.7
Hypopharynx	4	7.0	3	5.0
Larynx	13	22.8	21	35.0
Salivary gland	1	1.7	0	0.0
Other	6	10.5	3	5.0
More than one site	1	1.7	0	0.0
None identified	3	5.3	1	1.7
Disease status				
Previously untreated	1	1.8	3	5.0
Recurrent	56	98.2	57	95.0
Primary tumor status				
Eradicated	13	22.8	12	20.0
Eradicated but recurred locally	35	61.4	35	58.3
Residual disease	6	10.5	8	13.3
Untreated	1	1.8	4	6.7
Unknown	2	3.5	1	1.7
Metastatic disease status				
Metastatic disease	35	61.4	41	68.3
No metastatic disease	22	38.6	19	31.7

Hypomagnesemia was more common among patients on arm A than on arm B (14% v 0%; $P = .006$).

Among the 111 eligible patients who started therapy, overall hematologic toxicity, defined as at least one event of grade 3 or 4 leucopenia, neutropenia or thrombocytopenia, occurred in 36% of patients on arm A, as compared with 18% on arm B ($P = .04$). Logistic regression analysis was used to model the probability of grade 3/4 hematologic

Table 2. Grade 3 and 4 Toxicity (percent of patients)

	Cetuximab (n = 58)		Placebo (n = 58)	
	Grade 3	Grade 4	Grade 3	Grade 4
Allergic reaction	3	3	2	—
Inner ear/hearing	3	—	3	—
Hemoglobin	12	3	9	—
Neutrophils	28	2	12	2
Platelets	9	2	2	2
Transfusion RBCs	12	—	10	—
Syncope	7	—	2	—
Thrombosis/embolism	3	3	2	—
Fatigue	17	—	14	—
Rash/desquamation	16	—	—	—
Injection site reaction	2	—	—	—
Anorexia	10	2	2	3
Dehydration	12	—	10	—
Nausea	24	—	19	—
Vomiting	17	2	17	—
Infection	15	—	10	—
Hypokalemia	5	5	9	2
Hypomagnesemia	10	3	—	—
Hyponatremia	24	2	26	2
Neuropathy, sensory	3	3	2	—
Dyspnea/hypoxia	10	5	2	2

toxicity as a function of treatment arm, with the additional covariate of number of treatment cycles administered included in the model. The number of treatment cycles received is significantly associated with the probability of hematologic toxicity for a given group (odds ratio [OR] = 1.24; 95% CI, 1.03 to 1.48; $P = .02$). Controlling for the number of treatment cycles received, treatment arm is not a significant predictor of hematologic toxicity (OR = 2.14; 95% CI, 0.85 to 5.41; $P = .11$).

Toxicity data were available after cross-over for 14 of 15 patients on step 2. Grade 3 or higher toxicity was observed in 85% of patients on step 2. Grade 3 anemia was seen in 7%, and 14% received RBC transfusions on this step. Grade 4 neutropenia was reported in 14%, and high-grade fatigue, anorexia, dehydration, colitis, dysphagia, vomiting, hyperkalemia, hypokalemia, neuropathy, thromboembolism, and hallucinations each occurred in one patient on step 2. Two patients had grade 3 hyponatremia.

Skin toxicity is an expected manifestation of cetuximab therapy. Skin toxicity is defined as the presence of one or more incidents of rash/desquamation, dry skin, nail changes, or other skin toxicity of all grades according to the National Cancer Institute Common Toxicity Criteria version 2.0. Skin toxicity was reported for 43 of 56 eligible patients (77%) on the cetuximab-containing arm, compared with 24% of patients on the cisplatin and placebo arm ($P < .001$). Skin toxicity was grade 3 in 10 patients (23%). The development of rash was common enough that the use of a placebo was likely ineffective

in blinding patient and investigator to treatment assignment over time in many cases.

Hypersensitivity reactions were observed on both arms. Three percent of patients on the cisplatin and cetuximab arm had grade 3 hypersensitivity reactions, and 3% on this arm had grade 4 reactions. Those patients with grade 4 reactions were not re-treated with cetuximab study agent. Two percent of patients receiving cisplatin and placebo on step 1 had grade 3 hypersensitivity reactions, and there were no grade 4 reactions on this arm.

PFS and OS

At the time of this final report, 110 (94%) of 117 eligible and analyzable patients have died. No disease status information was submitted for one patient who survived for 13 months; this patient is included in the OS analysis but not analysis of PFS. Four patients (one on the cetuximab-containing arm and 3 on the placebo-containing arm) who died within 2 months of study entry, and on whom no information on disease status was submitted, are included as events at the time of death in the PFS analysis. Seven patients on arm A and four patients on arm B died without documented progression of disease. Among patients on arm A, two of these deaths were of unknown cause, two were related to the underlying disease, and three were related to neither treatment nor disease. Among patients on arm B, all four deaths were attributed to the underlying disease.

PFS was not statistically different between the treatment groups ($P = .09$), as shown in Figure 1. Median PFS was 2.7 months (95% CI, 1.9 to 3.8 months) for patients treated with cisplatin and placebo and 4.2 months (95% CI, 3.71 to 5.55 months) for patients treated with cisplatin and cetuximab. The corresponding comparison among all randomly assigned patients resulted in median PFS of 3.1 and 4.2 months, respectively, for the placebo-containing versus the cetuximab-containing arm ($P = .07$).

Survival was also not statistically different between the treatment groups, as shown in Figure 2. Median survival was 8.0 months (95% CI, 6.1 to 10.6 months) for the control

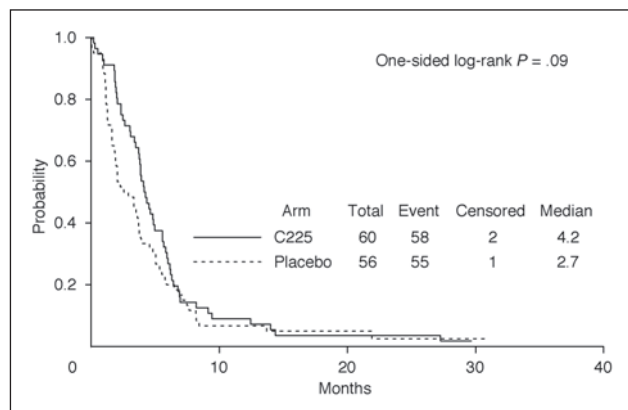


Fig 1. Progression-free survival by treatment group.

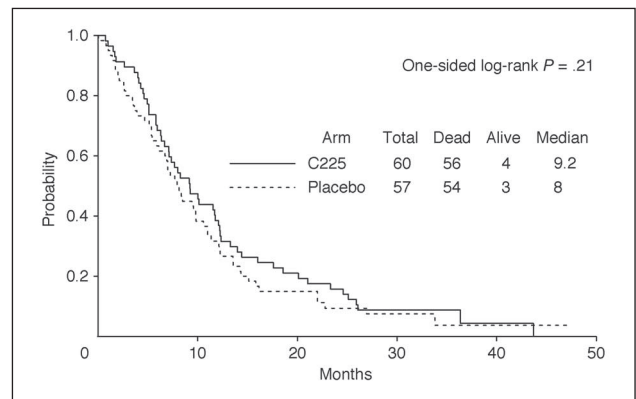


Fig 2. Overall survival by treatment group.

arm and 9.2 months (95% CI, 7.1 to 12.1 months) for the experimental arm ($P = .21$). The corresponding comparison among all randomly assigned patients resulted in median OS of 7.9 months and 9.2 months for the control and experimental arms, respectively ($P = .21$). The 1-year survival rates were 38.6% (95% CI, 26.0% to 51.2%) for patients assigned to cetuximab and 31.7% (95% CI, 19.9% to 43.4%) for patients assigned to the control arm. The 2-year survival rates were 15.8% (95% CI, 6.3% to 25.3%) for patients assigned to the cetuximab arm and 9.4% (95% CI, 1.8% to 16.9%) for patients assigned to the control arm.

The median PFS for patients on step 2, from the time of cross-over, was 3.5 months, and the median OS from the time of cross-over was 3.9 months. The median time to progression on step 1 for the 13 eligible patients who entered step 2 was 2.3 months (95% CI, 1.2 to 5.0 months). The median time on treatment for patients who entered step 2 (time on step 1 plus time on step 2) was 6.6 months (95% CI, 4.4 to 8.9 months). Individual log-rank tests comparing PFS and OS by important patient characteristics were conducted. Age group (categorized as < 55 years, 55 to 69 years, or ≥ 70 years), alcohol consumption, smoking status, sex, or weight loss were not significant predictors of PFS or OS ($P > .20$ in each case). Performance status (0 v 1) was a marginally significant predictor of OS ($P = .05$) but not PFS.

A Cox proportional hazards regression model for PFS comparing treatment groups confirms the log-rank test from above and estimates a hazard ratio of cisplatin with cetuximab to cisplatin with placebo of 0.78 (95% CI, 0.54 to 1.12). Proportional hazards regression modeling of PFS on treatment group, controlling for baseline characteristics, did not change the results compared with the analysis by treatment group alone. There were no interactions between treatment group and the prospectively defined baseline characteristics. Sample size differs substantially within various patient groups, and the power for conducting treatment group comparisons between subgroups was limited.

We examined survival as a function of skin toxicity because of suggestions from studies of cetuximab therapy in

colon cancer¹³ and of other EGFR inhibitors in head and neck cancer¹⁴ that the development of skin toxicity is a surrogate for biologic activity of EGFR inhibitors. Skin toxicity is an internal time-varying covariate: that is, patients develop skin toxicity at different points in time after the initiation of cetuximab therapy. Skin toxicity will be observed more frequently in patients who survive long enough to develop the toxicity. Cox regression modeling with a time-varying covariate was used to analyze the relationship between skin toxicity and survival. All OS and PFS analyses comparing skin toxicity groups among cetuximab-treated patients were restricted to those patients who did not experience disease progression within 1 month of entering study, representing approximately one cycle of therapy. Four patients on arm A experienced disease progression or died in the first month on study, of whom one patient had developed skin toxicity. Survival analyses by skin toxicity group concern the remaining 53 eligible patients on arm A.

Once a patient had developed skin toxicity of any grade at a given time *t*, the patient was considered to have skin toxicity at any time after *t*. Skin toxicity was observed in 42 (79%) of these 53 patients; at the time of development, this was grade 1 in 17 patients (40%), grade 2 in 20 patients (48%), and grade 3 in five patients (12%). The median time to development of skin toxicity for the 42 cetuximab-treated patients with reported skin toxicity was 1.1 months, with minimum of 0.3 and maximum of 3.7 months. Seventy-five percent of patients had developed skin toxicity by 1.7 months.

Because patients develop skin toxicity at different points in time after the initiation of cetuximab therapy, a Cox proportional hazards regression model was used with skin toxicity as a time-varying dichotomous covariate to permit inclusion of all skin toxicities and the actual times at which they were first reported. The estimated hazard ratio for survival by development of skin toxicity was 0.42 (95% CI, 0.21 to 0.86), with a statistically significant score test *P* value of .01. Thus the risk of death at any time *t* is 2.36-fold

higher for a patient without skin toxicity than for a patient with skin toxicity. Skin toxicity was not associated with PFS. In the proportional hazards regression model of PFS, the estimated hazard ratio was 0.74, with a *P* value of .37.

Response

There were 112 eligible patients assessable for response. The reasons patients were unassessable for response were as follows: patient died before commencing therapy (*n* = 2), no baseline disease assessment submitted (*n* = 1), and no repeat tumor assessments done (*n* = 2).

The objective response rate was 10% for patients receiving cisplatin with placebo and 26% for patients receiving cisplatin with cetuximab. This difference was statistically significant (*P* = .03). No responses were observed with cetuximab-containing therapy on step 2; nine (69%) of 13 patients had stable disease for at least 4 weeks, one patient (8%) had progressive disease, and repeat tumor assessment data were not submitted for three patients (23%). The response rate for the 43 patients on arm A who developed skin toxicity was 33% (14 responders), compared with 7% (one responder) among the 14 patients on arm A who did not develop skin toxicity. This difference was not statistically significant (*P* = .08), although the study was not designed to test this comparison and its power is low.

EGFR Expression

Tumor material for EGFR expression analysis was submitted for 99 patients. Informative samples were received for 86 eligible patients. Table 3 lists the EGFR staining results by treatment arm. The distribution of low-to-moderate and high EGFR immunoreactivity did not differ by treatment arm. There were 17 cases staining 3+ on ≥ 80% of cells among the 44 specimens (38.6%) stained for EGFR from patients on arm A, and 17 cases staining 3+ on ≥ 80% of cells among the 42 specimens (40.5%) stained for EGFR from patients on arm B. Sex did not differ between EGFR groups.

Table 3. EGFR IHC Results by Treatment Group

Table 2. EGFR IHC Results by Treatment Group										
EGFR Result	EGFR Percentage									Total
	0.10	0.20	0.30	0.40	0.50	0.60	0.70	0.80	0.90	
Cetuximab-treated patients										
1	0	0	3	0	0	0	0	0	0	3
2	0	1	2	1	0	0	0	0	0	4
3	5	0	5	1	3	3	3	4	13	37
Total	5	1	10	2	3	3	3	4	13	44
Placebo-treated patients										
1	0	0	0	0	0	0	0	0	0	0
2	2	0	1	0	0	0	0	0	0	3
3	3	2	4	1	4	3	5	8	9	39
Total	5	2	5	1	4	3	5	8	9	42

Abbreviations: EGFR, epidermal growth factor receptor; IHC, immunohistochemistry.

There was a marginally significant difference in response by EGFR expression status. The response rate was 27% (14 of 52 patients) for patients with EGFR low-to-moderate staining, whereas it was 9% (three of 34 patients) for patients with high EGFR staining intensity and density ($P = .05$). Subgroup analyses showed a statistically significant difference in response between treatment arms among EGFR low-to-moderate patients, but not among EGFR-high patients (Table 4). Among the 52 patients categorized as EGFR low-to-moderate, 12% (three of 25 patients) of those treated with cisplatin and placebo responded, compared with 41% (11 of 27 patients) of those treated with cisplatin plus cetuximab ($P = .03$). Among the 34 patients categorized as EGFR high, 6% (one of 17 patients) responded to cisplatin and placebo, versus 12% (two of 17 patients) who responded to cisplatin plus cetuximab ($P = .99$). In a logistic regression analysis of response, the interaction between EGFR and treatment group was not found to be statistically significant. However, we note that there was insufficient power to detect differences within and between subgroups based on these small sample sizes.

PFS and OS were not shown to differ by EGFR expression. Exploratory comparisons of treatment arms within EGFR subgroups for PFS and OS were not statistically significant. The median PFS and OS for EGFR-high patients randomly assigned to cisplatin with cetuximab were 3.9 and 6.7 months, compared with 3.6 ($P = .68$) and 9.8 months ($P = .45$) for EGFR-high patients randomly assigned to cisplatin and placebo. Median PFS and OS for EGFR low-to-moderate patients assigned to cisplatin and cetuximab were 5.0 and 10.0 months, compared with 3.3 ($P = .27$) and 7.0 months ($P = .32$) for those assigned to cisplatin and placebo. Cox proportional hazards regression modeling of PFS and OS to examine the relationship between EGFR status, treatment group, sex, and their interactions in survival analyses did not yield any significant results. The hazard rates for progression or death seemed not to be constant over time between the EGFR groups. For this reason, because of limited sample size for these analyses, and because of the second-line use of cetuximab in some patients, inferences based on the Cox modeling should be interpreted with caution.

Table 4. Objective Response

	% of Patients		<i>P</i>
	Cisplatin + Cetuximab	Cisplatin + Placebo	
Overall RR	26.3	9.8	.029
Low-to-moderate EGFR	40.0	11.5	.027
High EGFR	11.8	5.9	.99

Abbreviations: RR, response rate; EGFR, epidermal growth factor receptor.

There were 44 patients assigned to cisplatin and cetuximab for whom EGFR staining results were available. Skin toxicity developed in 22 (81%) of 27 patients with EGFR low-to-moderate cancers, compared with nine (53%) of 17 patients with EGFR-high cancers ($P = .09$).

DISCUSSION

This randomized, multicenter, placebo-controlled study of cisplatin monotherapy or cisplatin given together with cetuximab demonstrated that cetuximab is active in the first-line management of recurrent and metastatic squamous cell carcinoma of the head and neck. There was a significant increase in the objective response rate for patients treated with cisplatin and cetuximab relative to those who received cisplatin with placebo. The magnitude of this difference was comparable to the activity of cetuximab when given together with cisplatin in cisplatin-refractory head and neck cancer.^{15,16} Our results confirm the findings of prior studies that demonstrate poor prognosis for these patients. The median PFS observed in our control patients was 2.7 months, longer than the 2 months projected when the study was designed. This slight improvement over historical controls reflects the enrollment only of good performance status patients and may also be influenced by the widespread use of computed tomography imaging and advances in supportive care. The addition of cetuximab to cisplatin monotherapy reduced the risk of progression by 22%. This was not significant in a study powered to detect a 50% reduction in hazard rates; however, to detect a 2-month prolongation of median PFS from 2.7 months with 90% power, approximately 173 patients would have been required, rather than the 123 we enrolled.

As has been described previously for patients with advanced colorectal cancer and those with head and neck cancer treated with cetuximab or with the EGFR kinase inhibitor gefitinib, patients who developed the characteristic skin toxicity of cetuximab survived longer.^{13,14,16,17} The response rate for patients in arm A with skin toxicity was 33%, compared with 7% for patients who did not develop skin toxicity ($P = .08$). The power for this comparison was low.

Data regarding EGFR expression were analyzed by intensity and density of staining. A breakpoint of 3+ expression on $\geq 80\%$ of cells was chosen because of the apparent adverse prognostic impact of such high staining intensity and density; because 3+ staining is common in head and neck cancer and was so in our series as well, to separate out the subset with richest EGFR expression, a slightly higher immunoreactivity score was selected than in the reference experience in esophageal cancer, recognizing that EGFR immunoreactivity has a continuous relationship with outcome.^{18,19} The highest density and intensity of staining predicted for relative resistance to the addition of cetuximab. This

is the first demonstration that EGFR staining may predict for outcome in cetuximab-treated patients. The interesting observation that skin toxicity tended, albeit nonsignificantly in a small sample, to be less likely in patients who had the highest levels of immunohistochemical staining for EGFR raises the possibility that the cetuximab dose and schedule we used did not result in saturation of the large reservoir of EGFR molecules present in these patients so that less cetuximab was available to bind to EGFR present in the skin.

Investigators hypothesized that cetuximab would be most active in tumors with the highest levels of EGFR expression. Our finding that the activity of cetuximab is clearest in patients who do not have the highest receptor staining density and intensity was not anticipated. The small sample sizes for these comparisons were small, thus the finding may be erroneous. Other possible explanations are as follows: (1) the possible failure of cetuximab at the currently recommended doses to saturate the high number of EGFRs present when EGFR staining is 3+ on $\geq 80\%$ of cells; (2) at high receptor density, stochastic interactions or other ligand-independent mechanisms of EGFR activation

that cetuximab would not inhibit may be more common; or (3) constitutive activation of signal transduction molecules downstream of EGFR may be important determinants of cetuximab resistance that we did not measure.

A next step being pursued in the ECOG is the addition of cetuximab to chemoradiotherapy for locally advanced resectable or unresectable head and neck cancer. Future studies should examine EGFR expression with highly quantitative immunohistochemical methods, such as those using automated image analysis, to confirm or refute our finding that highest EGFR expression levels predict for relative resistance to cetuximab.²⁰ Treatment strategies to be explored for patients whose tumors express EGFR at the highest levels include the use of higher cetuximab doses, combinations of cetuximab with inhibitors of EGFR kinase activity, and combinations with agents that target signal transduction molecules downstream from EGFR.

Acknowledgment

We thank the 123 patients who consented to participate in this study, ECOG investigators, Elisa Manberg, Carol Getman, and Dr Richard Wheeler.

Authors' Disclosures of Potential Conflicts of Interest

Although all authors completed the disclosure declaration, the following authors or their immediate family members indicated a financial interest. No conflict exists for drugs or devices used in a study if they are not being evaluated as part of the investigation. For a detailed description of the disclosure categories, or for more information about ASCO's conflict of interest policy, please refer to the Author Disclosure Declaration and the Disclosures of Potential Conflicts of Interest section in Information for Contributors.

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Dollar Amount Codes (A) < \$10,000 (B) \$10,000-99,999 (C) $\geq$ \$100,000 (N/R) Not Required								

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ERRATA

The October 1, 2005, article by Romaguera et al titled, “High Rate of Durable Remissions After Treatment of Newly Diagnosed Aggressive Mantle-Cell Lymphoma With Rituximab Plus Hyper-CVAD Alternating With Rituximab Plus High-Dose Methotrexate and Cytarabine” (J Clin Oncol 23:7013-7023, 2005) contained inaccurate information in Table 4 as follows:

Whereas the overall 3-year FFS was given as 65%, it should have read 64%.

Whereas the 3-year FFS of age > 65 years (No) was given as 75%, it should have read 73%.

Whereas the *P* value for age > 65 years (No) was given as .01, it should have read .02.

Whereas the 3-year OS for age > 65 years (No) was given as 85%, it should have read 86%.

Whereas the 3-year OS for age > 65 years (Yes) was given as 75%, it should have read 74%.

DOI: 10.1200/JCO.2006.12.002

The November 20, 2005, article by Carli et al titled, “Pediatric Malignant Peripheral Nerve Sheath Tumor: The Italian and German Soft Tissue Sarcoma Cooperative Group” (J Clin Oncol 23:8422-8430, 2005) contained inaccurate information.

The spelling of the tenth co-author’s name was given as Eura Koscielniak, whereas it should have read Ewa Koscielniak. The Authors’ Disclosure of Potential Conflicts of Interest section should have contained the following statement: “The authors indicated no potential conflicts of interest.”

DOI: 10.1200/JCO.2006.12.001

The December 1, 2005, article by Burtneß et al titled, “Phase III Randomized Trial of Cisplatin Plus Placebo Compared With Cisplatin Plus Cetuximab in Metastatic/Recurrent Head and Neck Cancer: An Eastern Cooperative Oncology Group Study” (J Clin Oncol 23:8646-8654, 2005) contained incorrect information.

In the Abstract, in the second-to-last sentence of the Results section, the values for objective response rate were interchanged. Whereas 10% was given for arm A and 26% for arm B, it should have read 26% for arm A and 10% for arm B.

In Figure 1, the median values were interchanged. Whereas 2.7 was given for the C225 arm and 4.2 for the placebo arm, it should have read 4.2 for the C225 arm and 2.7 for the placebo arm.

In Figure 2, the median values were interchanged. Whereas 8 was given for the C225 arm and 9.2 for the placebo arm, it should have read 9.2 for the C225 arm and 8 for the placebo arm.

The online version has been corrected in departure from the print.

DOI: 10.1200/JCO.2006.12.003
